INFO162 - Introduction to Human Computer Interaction
Fall 2023
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Into the Mind. An app to help

Project report - Human Computer Interaction (INFO162), University of Bergen

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1. Vision

1.1. The problem space

The problem space is about the needs of the users, and defining a problem space involves finding out how the product will improve the user experience (Sharp et al. 2019, s.43). When designing a product, you need to know of the flaws in the existing products with the same target group (Sharp et al., 2019, s.72). We have decided to look into problems with the existing mental health institutions, specifically therapy in Norway. We will also be looking into websites with similar target groups.

If you get an appointment with a psychologist without a referral from your primary care physician, you will have to pay this cost in full. The cost of a psychologist can vary, but is in general considered expensive (Eliassen, 2022). On average 30 % of people referred to the mental health sector by their primary care physician will be rejected (Eliassen, 2021). This means that 30 % of people the doctors think need therapy will not get it from the public sector. We see here two big problems with therapy in Norway, firstly it is difficult to get an appointment, and secondly it can be very expensive. We hope our product can offer solutions to these problems.

1.2. Target Group

The app is directed at people suffering from mild, pre-clinical mental health issues, like motivational problems/pre-clinical depression, feeling alone or going through a tough time. With this app they could get that extra support and understanding of what they are going through (surveys, chat with doctors, mindfulness games or routines, etc.), as well as meeting new people that might be in the same situation as them to help each other to understand their situation better and improve how they feel.

1.3. Similar Work

There is a website called *Moodgym* that offers a free-of-charge online course for your mental health. It was developed by the Australian National University and consists of the modules Feelings, Thoughts, Unwarping, Destressing, Relationships, Workbook. It regularly offers to test your own mental health in different areas like depression and fear (ehub Health, 2023).

As an online course, it doesn't require users to book an appointment, making for a low entry bar which is especially important for both users fighting with low motivation and for users who are not sure whether their mental problems are bad enough to warrant help from an expert. Additionally, it asks its users questions about their age, mental state etc., and then personalises the course to their situation. In order to make it relatable, it includes fictional characters of a similar age as the users who have or don't have different mental problems. Another positive characteristic is that it was developed by mental health experts, and it was scientifically proven to be effective (ehub Health, 2023).

On the other hand, after a few weeks the course is done, so presumably the positive effects will fade some time after the completion. Moreover, since the course is implemented as a website, it can't send notifications or reminders to the user and the hurdle to open it is higher than an app, so especially users fighting with low motivation or pre-clinical depression might not manage to complete the course.

There is a Norwegian app called *Wake me up Norway* that has a similar concept as our app. They offer video calls with coaches and mentors, lectures and workshops, and the option to join a group of similar minded people. These features are entirely free, and the app relies on sponsors for money (Wake me up Norway, u.å). One of our concerns is what happens if the sponsors stop supplying money? Will the app still function? Furthermore, for some people it might feel like too much of a hurdle to book an appointment, or reach out to mental health professionals. We would like for our app to have a very low bar, so no one might feel disinclined to use it.

1.4. Solution

We want to design an app to help people who suffer from mild mental health issues. While an app can’t replace psychotherapy, it can be the first place to go with a really low entry bar. It can help people who are waiting for an appointment with their therapist, and it can help people who just can’t afford therapy.

Our app helps people with mental health issues and their loved ones. We would like to have functionalities like: surveys to check in on the user’s current mental state, reminders for the user to take care of themselves and their friends, and a chat function that can connect users to each other. Our product would be free of charge, and available at all times. Because anyone can access the app from their phone, there will be a low bar for users to use the app.

The first idea for the project was coming up with a way to improve some aspects of the health and wellbeing field. We went through several ideas from improving the way you could get medicine and quicker doctor appointments to some type of scanners that could detect diseases. In the end the group decided to focus more on mental health which is a topic that resonated better with us.

2. Requirements specification

2.1. Data gathering

We chose to use a digital questionnaire, since it is completely anonymous, while in-person methods like interviews could deter people from opening up (Sharp et al. 2019, s. 278-286). Our goals are finding out how we should prioritise different mental health problems like loneliness, slight depression, and insecurity; finding out which percentage of users would consider installing an app; and which features users would like. We shared the questionnaire with our fellow students. As we want anyone of any age and gender to use our app, we did not think it relevant to ask the participants for age or gender. This does, however, limit our

means of analysing the data. When we later create a persona, the age and gender is thus based on our presumptions and not any actual data. If we did further research, we would include more questions regarding the status of the person answering.

2.2. Survey Development

In order to achieve an effective survey, we researched other popular and contrasted surveys that had to do with our topic. One of the most interesting ones we found and took inspiration from is the R-UCLA survey (Russel et al., 1980). This survey is the standard measure of loneliness. The R-UCLA was designed to be self-administered; it has 20 items with four response categories each. We have chosen only a few since we wanted to cover other areas that R-UCLA didn't, like personal motivation.

We used a combination of open and close ended questions as open ended questions could suggest new features and issues we hadn't thought of, and we still keep close-ended questions that are easier to evaluate in order to create a persona.

We wanted to keep the survey as short as we could so the users that took the survey did not get tired whilst doing it and end up not giving accurate answers because they just want to get it over with.

2.3. Results

We got 37 answers in total. Since we sent the questionnaire to fellow students, we assume most of them had an academic background and were about age 20-30.

In order to analyse these responses we decided to plot them using bar graphs. We made sure to remove incongruent answers and responses that seemed unreal. We also have omitted some outliers.

The answers to the first two questions, shown in figure 1 and 2, told us that there are, in fact, issues with loneliness in this group.

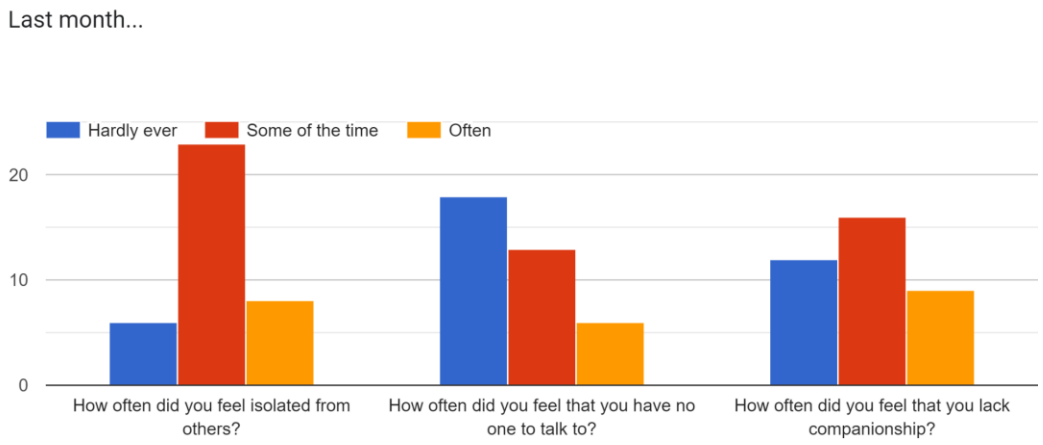

Figure 1

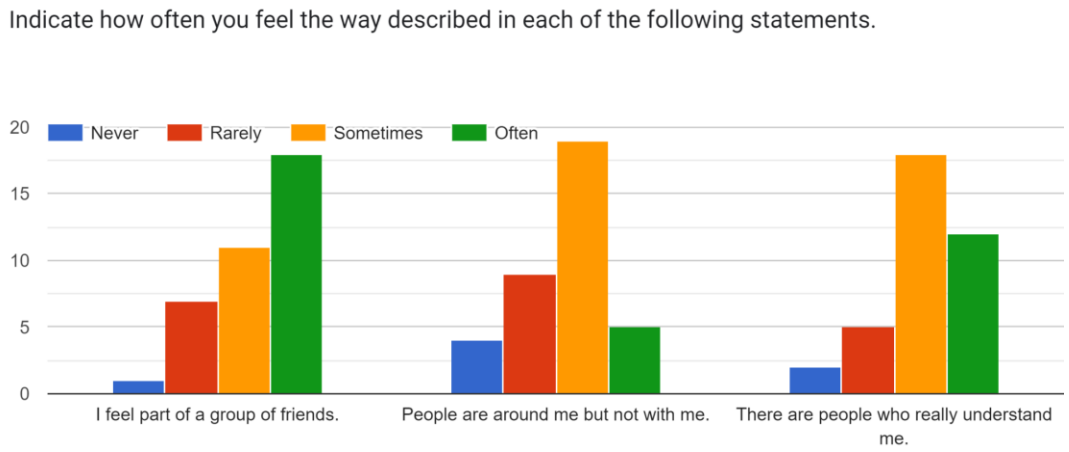

Figure 2

The third question, shown in figure 3, told us that on average, there are slightly more problems with feeling that one is not enough (average 3.16) than feeling mildly depressed (average 2.35) or feeling lonely (average 2.76), but the difference is minor.

We performed pairwise sign tests (Independent Media Associates Inc, 2023) between the three kinds of mental problems, in order to know whether one of them is significantly more frequent than the others. Because this means 6 tests in total (A>B, A<B, A>C, A<C, B>C, B<C), we chose to use a rather small alpha of 1% for each test; this leaves us with a chance of 5.9% that we will make a false-positive if there is no difference between the different mental problems (0.99^6 = 0.941, 1-0.941 = 5.9%). Table 1 shows n the results, with A=feeling not enough, B=mildly depressed, C=lonely.

	A-B	**B-C**	**A-C**
“≠0”:	26	19	22
“>0”:	21	5	15
“<0”:	5	14	7
P-value	0.00098	0.022	0.041
P-value < alpha	Yes	No	No

Table 1

The result is that there are more people who rated “Feeling like I am not enough” higher than “Feeling mildly depressed” than vice versa, which is statistically significant. We did not find any statistically significant differences in the other pairwise comparisons.

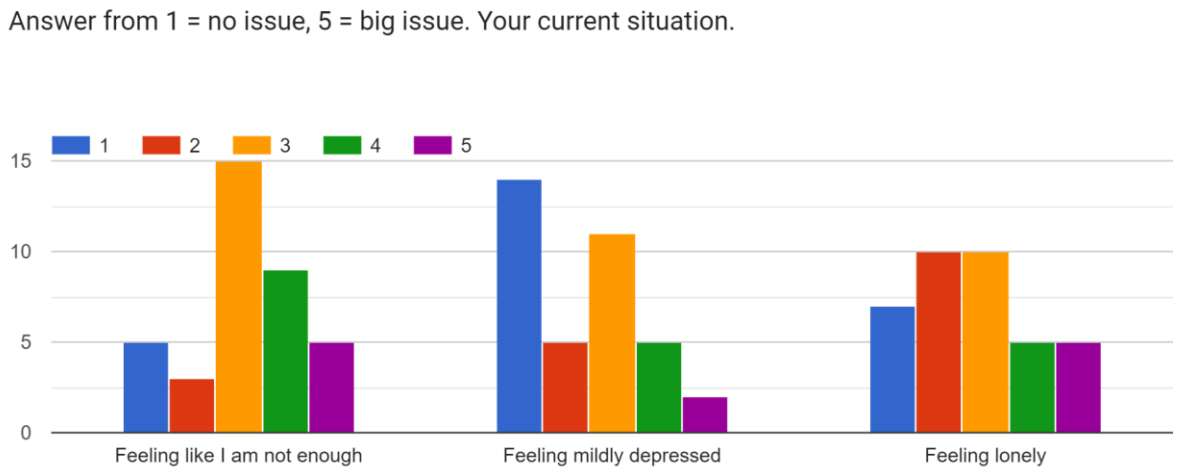

Figure 3

In the fourth question, see figure 4, only one of the participants would very likely download an app to help with mental health, but 10 participants answered 4/5 on the Likert scale and 8 answered 3/5, so we assume that there is a demand for such an app. Participants were instructed that 1 is not very likely and 5 is very likely:

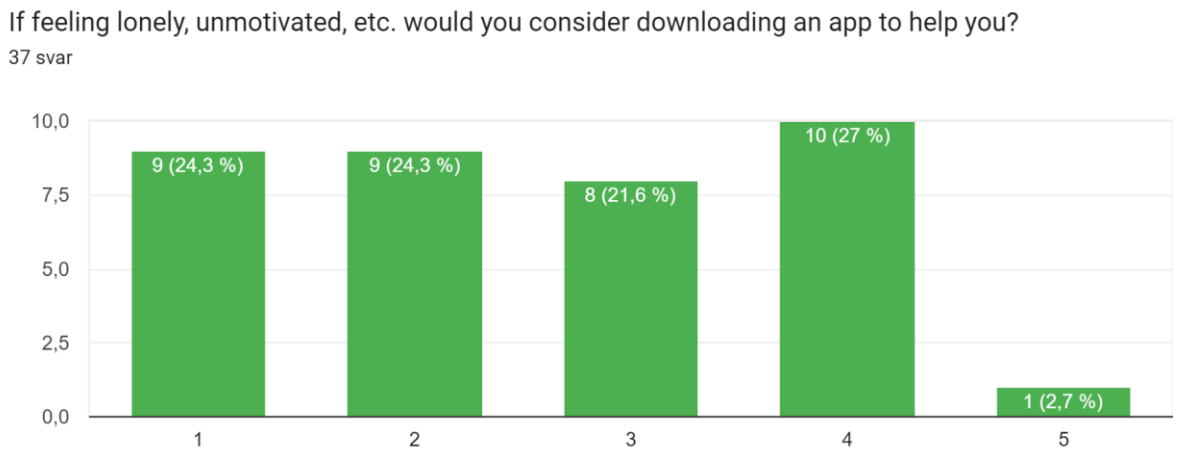

Figure 4

2.3.1. Open-ended questions

In question 1 we look for features that are lacking from existing mental health apps so we can find patterns of what users would like the existing apps to have; in the second we directly ask the users for ideas so we can find popular requests for features to develop in our app.

1. *Regarding already existing mental health resources. Which aspects do you think most of them often miss or don't exist at all*

Out of 16 answers, 3 asked for a way to socialize with others. Apart from that, we were not able to find any patterns in the answers.

2. *Which features would you like for a mental health platform to implement?*
   Out of 13 answers,
   - 4 wished for a way of interactivity/chatting
   - 3 wished for online therapies/exercises (for coping mechanisms, against panic attacks, and for quizzes to understand yourself better)
   - 2 wished for a way to contact a professional
   - 2 wished for anonymity
   - 1 wished for daily check ins

In summary, chatting and socialising with others is by far the most-requested feature.

2.4. Final requirements

Among others there are functional, non-functional and data requirements (Sharp et al., (2019), s.390).

2.4.1. Functional requirements

Our app needs a feature to *chat with other, like-minded users* because our data shows that many users wish for this. Since this will make it possible for users to harass and try to scam others, we will need an *effective way to report users*, which will then be blocked from using the app.

Additionally, it needs to *send reminders* in order to prevent users from forgetting about it.

2.4.2. Nonfunctional requirements

The app should *work on both Android and iOS*, which will cover ~99.3% of all mobile devices (Statcounter, 2023).

2.4.3. Data requirements

If the user wants to chat and socialise with others, the product needs to access *the user's phone number*: Firstly, in order to prevent them from logging in again after they were blocked for harassing or scamming others. Secondly, if someone successfully used our chat functionality to scam our users, we want to be able to give their identity to the police. It will also let a user access the app even if they change phones.

2.5. Persona

A persona is a description of a typical user, based on the data gathered. They should be described through their goals and frustrations, which are based on the requirements made from the data (Sharp et al., 2019, s.403). We have made a persona which we feel is a good representation of who might use our product. To create the persona we used a website recommended from *Interaction Design : beyond human-computer interaction*, called xtensio.

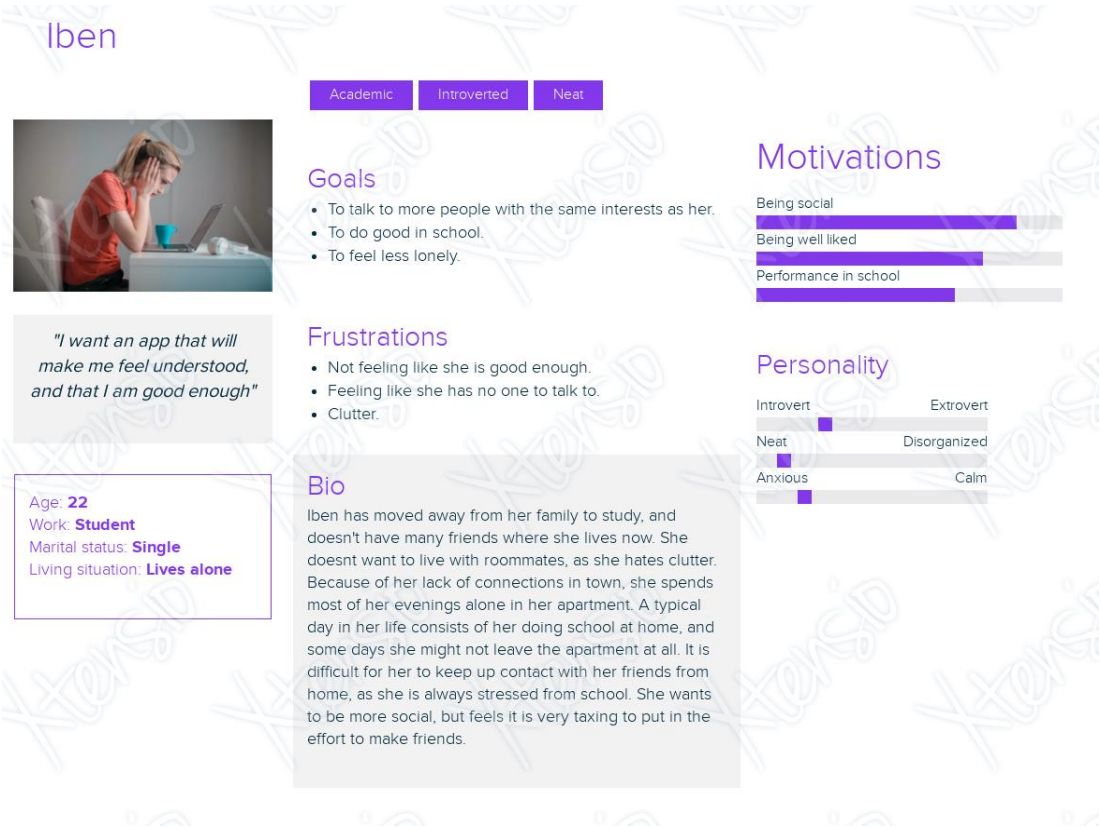

Figure 5 : Persona profile developed from analysing the results from the survey

3. Prototype description

We developed a storyboard and a low-fidelity prototype in order to show how our product is going to look and how users will interact with it (Sharp et al., 2019, s.422-428).

The storyboard will take you through two of the main functions inside the app, which are chat and reminders. It will show how a normal sequence with an actual user would develop in a normal case.

The prototype takes you through the whole app, it shows all the main screenshots of what the plan for the application is. From the screenshots you can get an idea of how the app buttons function and where they take you, but, it is not the main objective of the prototype.

3.1. Storyboard

With the storyboards, we want to show a fair estimation of what the app could be used as, without overselling it. E.g. in the beginning Iben isn't fully sad, she is just lonely, and in the end she isn't fully happy, but she feels less alone and has a slight smile on her face. We decided to use our persona in the storyboards, to make the situation feel more realistic, and to make it easier for us to create the story. To show some of the different functionalities in the app, we decided to make two storyboards, shown in figure 6 and 7.

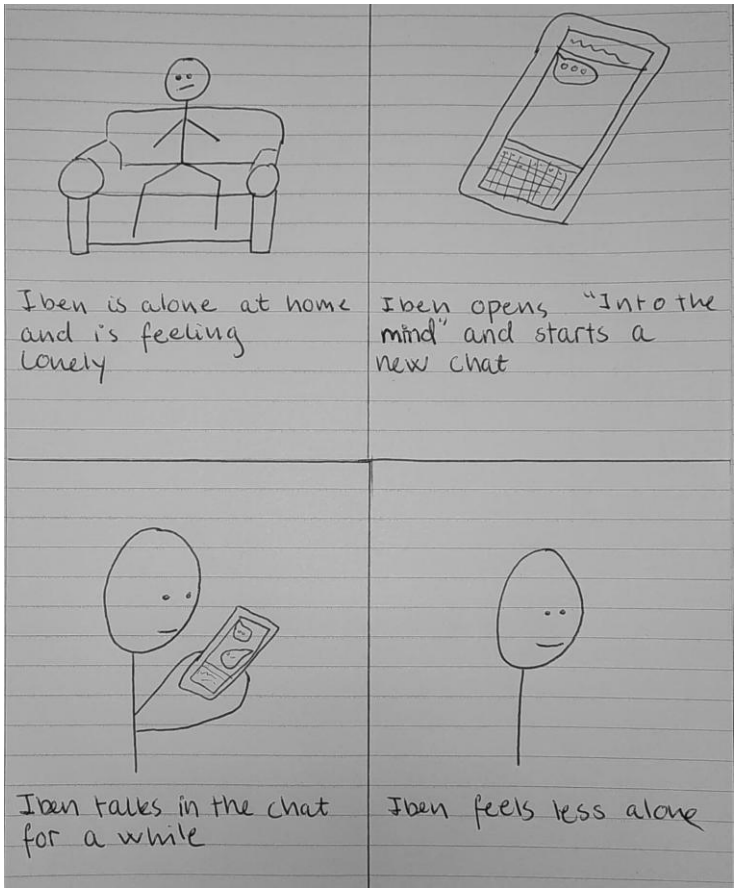

Figure 6 : Storyboard for chat function inside the prototype

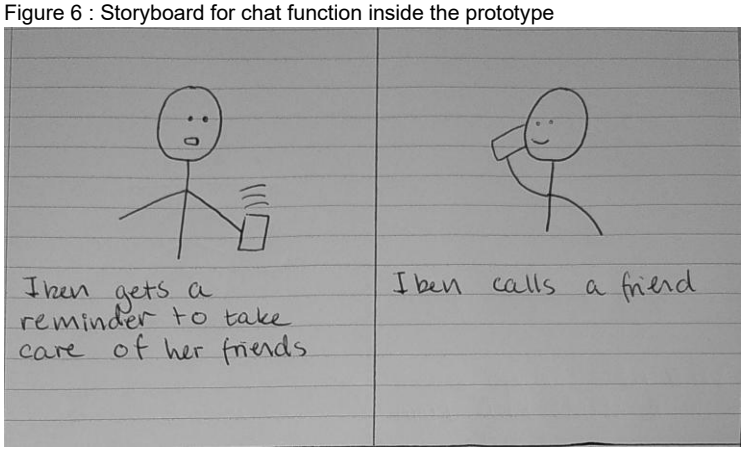

Figure 7: Storyboard for the reminders function inside the prototype

3.2. Low-fidelity prototype

We have developed the prototype using Balsamiq, a low-fidelity wireframing software that the group has experience with and made it easier to develop the low-fidelity prototype. Taking into account all the requirements for a low-fidelity prototype we developed the prototype explained and shown in the figure 8.

The start screen shows the app title and only 3 buttons, so that the user won't be overwhelmed by the number of options:

“Take mental health course”. This will allow users to take a purely online course, just like the existing website “Moodgym”, but directly inside the app instead of as a website (ehub Health, 2023). Shown in screen 5 of figure 8.

“Configure Reminders”. The app will send notifications to the users like “Say thanks to someone today” or to resume the mental health course. This button will allow the users to configure how often they appear, and what kind of notifications they want. We chose to call it “Configure Reminders” instead of “Configure Notifications” because we assume that “Notifications” is generally more associated with the phone bothering you, while “Reminders” is more closely related to the user’s be-goal of being reminded about mental-health practices. Shown in screen 4 of figure 8.

“Connect with like-minded people”. When a user clicks on this button, the first thing they need to do is log in with their phone number. Logging in with their number will also allow them to keep their settings and mental-health-course progress, but we decided to call the button “Connect with like-minded people” instead of “Log in with your number” so that users know which specific goal in the near future they can reach by logging in. Shown in screen 3 of figure 8.

In order for the design to give a calm impression, we chose to use rounded shapes for the buttons, no black shadows behind the buttons, and a mild background colour. We chose a yellowish green colour, because it produces positive expectations in certain consumer products (Won & Westland, 2018, s.105).

Figure 8 : Low-fidelity prototype developed taking into account requirements and main features

3.3. Design methods and steps

First of all we developed some very conceptual sketches on paper, mostly getting down the layout, what options we had for the buttons, which colours we wanted to use, etc.

After this we made a rough view on paper of how all the elements combined would look. From here we could evaluate the distribution of the elements on the screen and if we were happy with how it was looking. Lastly we passed all the information into the computer and developed the final low-fidelity prototype, with all the necessary elements already placed on the screenshot and ready to be evaluated.

4. Evaluation of the prototype

4.1. EVALUATION METHOD 1: Nielsen's Heuristics

To evaluate our prototype's usability, we chose Nielsen's Heuristics, which are 10 principles to check a product's compliance with (Nielsen, 1994). It allows for an evaluation method that is distinct from presenting the prototype to users and having them fill out a questionnaire, which allows us to get a different viewpoint than in the UX evaluation. We have chosen Nielsen's heuristics as our first method of evaluation primarily because it doesn't require users (Sharp et al., 2019, s.550). Most people have never used a mental health app before, and it can be difficult for them to test the design. The evaluation was done by 2 of us.

We didn't evaluate rules #5, #9, and #10 because our low-fidelity prototype doesn't contain any error messages, help or documentation. Furthermore, we didn't evaluate rule #7 because our app is not complex enough that shortcuts could significantly speed up the usage. Finally, #2 and #6 simply didn't apply to the part of our app we prototyped. Our low-fidelity prototype complies with the chosen rules as follows:

#1: Visibility of system status
Our app has three types of status: (i) Where in the app the user currently is, (ii) whether the user is signed in, and (iii) whether there is a working internet connection for the chat feature.
(i) is answered by the subtitles "Chat", "Reminders", and "Lesson [number]", while the title "Into the Mind" stays at all times so that the user can see which app they are currently using. The only possibly confusing part here is that the sign-up screen could be mistaken for a home screen if users don't see the grey "X" in the upper left part of the screen.
(ii) is answered by the text "(Necessary sign-in)" in the home screen.
(iii) is not visible inside the app right now. Since the operating system usually shows this status, it is debatable whether it is necessary; WhatsApp as a very popular app whose sole purpose is chatting also doesn't show this status.

#3: User control and freedom
Every screen of our prototype has a "Back"/"Close" button in the upper left corner, allowing users to go back after an unwanted tap.

#4: Consistency and standards
Most users never used a mental health app, removing the possibility to design our app similarly to similar products. It is however designed similarly to other apps in general. The process of signing in with a phone number is known to many users, and the layout of the chat and the reminders settings look the same as they do in most other apps.

#8: Aesthetic and minimalist design
Our prototype contains mainly the essential elements. Still, there are two not completely essential elements: The text “By signing up, you agree with the Terms of Service and Privacy Policy” isn’t necessary on the home screen and could be shown in the sign-up screen instead, where users would see it less frequently. And the sign-up screen doesn’t necessarily need a button “Need an account?”, because the user could be led to account creation automatically if no account is found after receiving a verification code.

4.2. EVALUATION METHOD 2: Questionnaire

Heuristics are a good method for starting the evaluation, however, it is not a sufficient replacement for user testing (Sharp et al., 2019, s.556). Experts using heuristics often miss severe problems, and tend to make problems out of nothing (Sharp et al., 2019, s.556). The primary goal of usability testing is to determine whether the intended target group can use the product for its intended purpose (Sharp et al., 2019, s.501). We decided to make an online questionnaire with questions that mirrors a walk through. Hopefully using the questionnaire gives us a bigger reach and more data to work with.

Results from questionnaire:
6 participants answered our questionnaire. It is not guaranteed that they are in our target group, as we didn’t add questions guaranteeing this. Our current data can give us a feel for how to proceed with the design, including what areas of the app needs reevaluation, but cannot be used to make any conclusive decisions.

1.

Figure 9: visual representation of question 1.

One of those who answered no elaborated with “it seems too risky sharing that kind of information after years and years of "New shocking data breaches/leakages"”.

2.

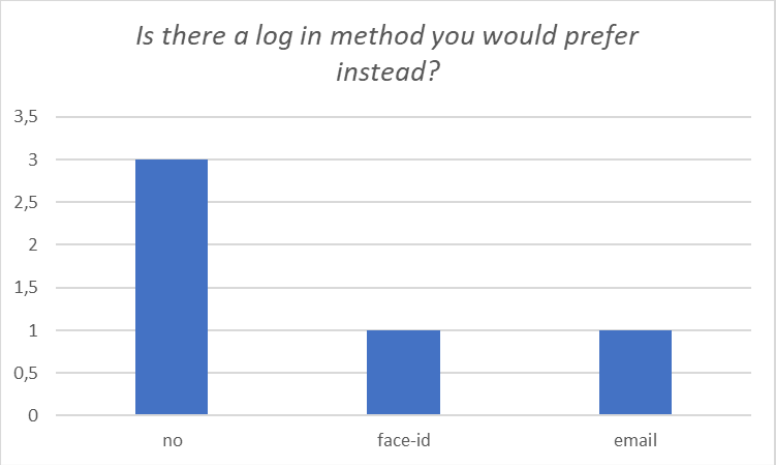

Figure 10: visual representation of question 2.

3. *From this start screen, please describe what you think the functionality of each section (the white boxes) are.*
   All the participants described the first functionality, take mental health course, as some variant of “take a course for your mental health”
   Most of the participants assumed the second functionality, configure reminders, is a way to set your notifications.
   Most participants described the last functionality as some version of “connect with other users”, but one participant described it as “connect with phone number or other social media like Facebook”.

4.

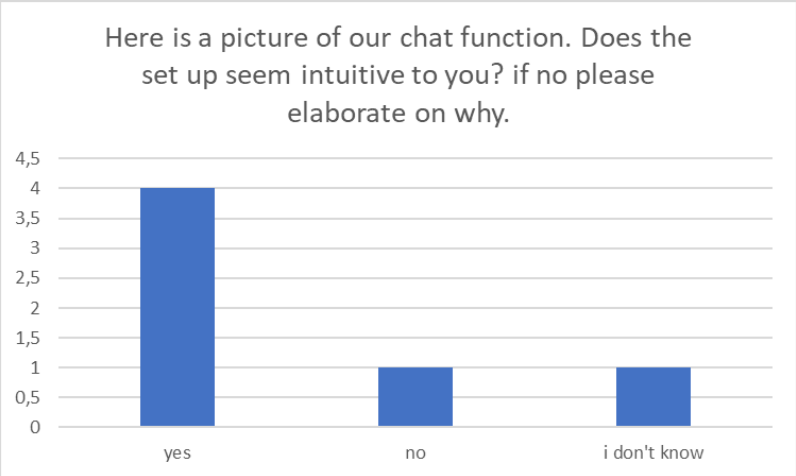

Figure 11: visual representation of question 4.

The person who answered no elaborated with “hard to read on phone”.

5.

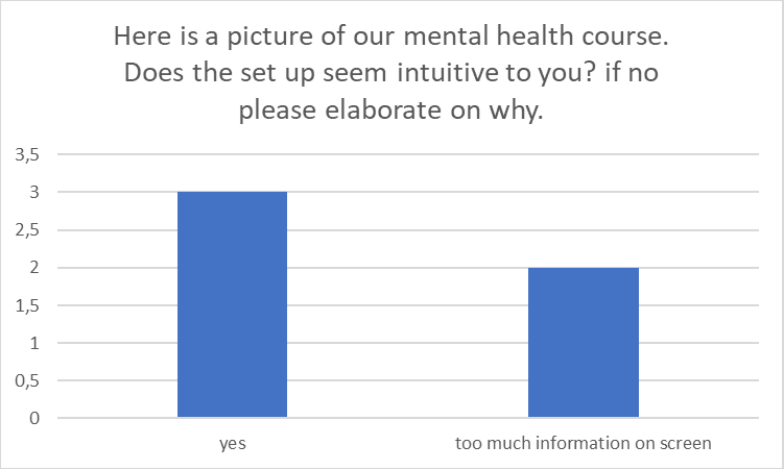

Figure 12: visual representation of question 5.

6.

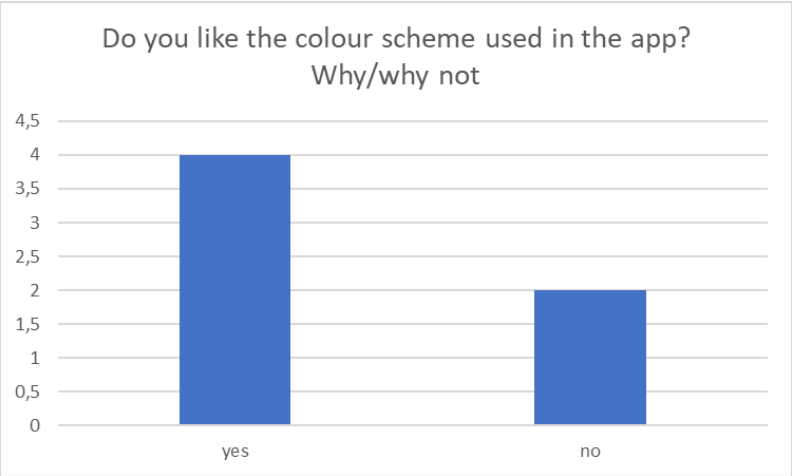

Figure 13: visual representation of question 6.

One who answered no elaborated with “it’s a weird green colour”, and one elaborated saying “The colour is not too relaxing as I would expect from an app made to improve mental health.”.

From the results, we see that it is good to not force the participants to log in when they don’t need to. We could do further research to see what method of log in the users would prefer, in order for them to feel comfortable logging in. When we further develop the mental health course site, we should take in mind the criticism regarding the amount of information on screen. We should then make sure to only have relevant information on screen at all times. In the future, we can reevaluate the green colour used in the app, and perhaps change it out for a different, more calming, green.

5. Discussion

5.1. Interpretation of evaluation results

Most of the evaluation's criticisms can be resolved in a straightforward way, so we are confident that we are overall on a good way. The one concern that may not have a good solution at all is the obligation to use your phone number to log in, for the reasons given in the next section.

5.2. Limitations of design process and evaluation

Our methods of evaluating the usability and user experience were not fit to uncover possible more fundamental problems with our idea. Our biggest concern is that the chat feature can lead to harassment or scams. The former could occur because we are connecting strangers with psychological problems, which might be serious enough that they harass others instead of helping each other or trying to make friends with them. The latter could occur because scammers might find lonely users of a mental health application an easy target.

In order to mitigate both problems, we decided to make it necessary to sign up with your phone number. This unfortunately makes it necessary for us to handle more personal data of our users and it increases the bar to use the chat feature. However, we see no other way to effectively block users and keep them from immediately signing up again. Also, it could make it easier to report scams to the police, at least if the phone numbers are from a country which works together with the Norwegian police, since phone numbers can identify the scammers.

Still, requiring phone numbers might not be enough. When launching our service, we will need to very actively monitor harassment and scams, and possibly take further measures like scam filters, because if we do not manage to effectively get this problem under control, this might force us to shut down our service.

5.3. Ethical Concerns

Talking about mental illnesses can stir up bad memories, so while gathering data, we made sure not to touch too sensitive topics like suicide. Users also didn't need to talk about other sensitive topics like loneliness, but needed only choose a value on a likert scale. By choosing to gather data with an online questionnaire, we afford users some anonymity.

Our app is going to handle health data, which are generally considered sensitive, so we will have to carefully consider which of these data we send over the internet. In order to comply with the GDPR, we will need to only send necessary data, include a privacy notice, and allow users to access, correct, and delete data we stored (European Union, 2023).

Furthermore, our app could be misunderstood as a replacement for psychotherapy by users with serious mental health issues, so we need to make clear that users should seek professional help if necessary.

5.4. Conclusion

We developed a vision for an app that has the potential to make a small improvement to people's lives by improving their mental health. Then we analysed the problem space by gathering data about the need and user's wishes for such an app, which we synthesised into requirements and a persona. We subsequently developed a prototype, which we evaluated using two different methods with mostly positive results. To test usability, we used Nielsen's heuristics, and to test user experience we used an online questionnaire.

The next step is to develop an alternative, improved design taking into account the evaluation results, and create a more detailed prototype from it, possibly already a high-fidelity prototype.

6. Individual Contributions

Candidate 178's contribution: Contributing to the Statistics and Motivations slide on the first presentation. Writing in concept and the problem space part of the report. Writing about wake me up Norway. Researching the problem space. Write about data gathering. Developing and writing about the persona. Developing the storyboards. Write in the storyboard section. Develop the evaluation questionnaire, write the evaluation method 2 section of the report.
Candidate 114's contribution: Developing the target group paragraph and introducing the pitch selection process on the report. Contributing to the Statistics and Motivations slide on the first presentation. Developing the survey. Researching other popular surveys about the topic such as the R-UCLA loneliness scale. Develop the low-fidelity prototype.
Fixing feedback from assignment 3. In the presentation, developing and presenting Prototype and Storyboard section. Plan for proceeding design. Restructuring the vision, and adding information to survey development.
Candidate 158's contribution: Coming up with the general idea of doing something for mental health, designing and presenting the last two slides of the presentation, writing the first version of the "Statistics and Motivations" part of the report, researching and writing about Moodgym. Develop the survey. Analyze the open-ended questions, write about functional and data requirements. Help with developing the low-fidelity prototype and storyboard, and write the report about them. Evaluate the usability using Nielsen's Heuristics. Writing the "Discussion" part of the report.

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